Elias Arellano Campos

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Work Authorization: U.S. OPT Active | TN Visa Eligible (Does not require H-1B sponsorship)

Education

University of Houston May 2026
B.S. in Computer Science, Minor in Mathematics Houston, TX
Concentration: Data Science & Machine Learning
Honors: Dean's List, Magna Cum Laude

Experience

Personal Projects & Applied ML/AI Work 2024 -- 2025

- Designed and deployed end-to-end GenAI applications integrating LLM APIs with custom retrieval pipelines, prompt engineering, and context-grounded generation.
- Built and productionized deep learning systems processing large-scale datasets, from architecture design through evaluation and REST API deployment.
- Enabled real-time model inference by integrating ML systems with Flask REST APIs and interactive dashboards (Streamlit), reducing time-to-insight for end users.
- Applied explainability techniques (SHAP, saliency maps) to validate model behavior and identify failure modes across diverse datasets.

Projects

EEG Seizure Detection *PyTorch, Python, MNE, CHB-MIT*

- Achieved best **F1 of 0.8537** and **AUC of 0.9924** by designing and comparing four deep learning architectures (CNN, CNN+LSTM, CNN+GRU, TCN) on **23-channel** pediatric EEG data.
- Addressed severe class imbalance (∼**0.23%** positive rate) using weighted loss functions, threshold optimization, and learning rate scheduling with gradient clipping.
- Implemented saliency map explainability and per-patient error analysis to identify generalization gaps and guide model improvement.

RAG Document Q&A Agent *Python, JavaScript, Claude API, pdf.js*

- Built an end-to-end RAG pipeline with TF-IDF indexing and cosine similarity retrieval supporting plain text and PDF ingestion via pdf.js.
- Integrated Claude LLM API for context-grounded answer generation with source-attributed chunk retrieval and similarity scoring.
- Implemented configurable chunking strategy with overlap to prevent context fragmentation, improving answer coherence across long documents.

Fraud Detection System *Python, scikit-learn, SHAP, pandas*

- Achieved **97.1% F1**, **98.2% recall**, and **96.0% precision** on the fraud class by training a Random Forest model on a **108K-sample** balanced holdout dataset.
- Reduced false positives to **84 FP** against **2,016** correctly identified fraud cases, with **0.999** ROC-AUC and **99.55%** overall accuracy.
- Increased model transparency by implementing SHAP-based explanations for both global feature importance and individual prediction interpretation.

Amazon Review Opinion Search Engine *Python, SBERT, NLP, Flask*

- Increased average precision by **47.7%** (**0.39** → **0.58**) while reducing retrieved document volume by **83%** by replacing keyword search with SBERT-based semantic retrieval.
- Cleaned and transformed **210K+** reviews using custom preprocessing pipelines; demonstrated cleaner data outperforms larger unfiltered sets.
- Deployed as a Flask API enabling real-time semantic query over the full review corpus.

Activities

- **Code Coogs**, University of Houston --- Participated in ML and AI workshops and collaborative technical sessions.
- **University of Houston Tennis Club** --- Competed in club matches; developed teamwork, discipline, and leadership.

Skills

- **Programming:** Python, R, SQL, C++, Java
- **ML & AI:** PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, pandas, SHAP, NLP, Deep Learning, GenAI, LLM APIs, Prompt Engineering, RAG, REST APIs
- **Computer Vision:** OpenCV, MediaPipe
- **Web & Tools:** Flask, Streamlit, Shiny, Git, Linux, Jupyter, VS Code
- **Languages:** English (Fluent), Spanish (Native)